

Efficient Text-to-Music Generation via Flow Matching with Bidirectional Mamba SSM

Anthony Wang (University of California, San Diego)

I. Grand Challenge Overview

Challenge Description

- ICME 2026 Grand Challenge on **Academic** Text-to-Music Generation
 - Why academic? Most SOTA text-to-music generators like MusicGen are models created in “big tech” companies
 - Trained on propriety datasets and have access to substantial compute
- This challenge: standardized, publicly available dataset (MTG-Jamendo) with no augmentation allowed
- Two tracks: efficiency ($\leq 500\text{M}$ parameters), performance (no parameter restrictions)

Challenge Description

- Objective Evaluation metrics: Fréchet Audio Distance (FAD), CLAP score, and Concept Coverage Score (CCS)
 - FAD: measures distributional similarity between generated music and real music. The lower this score is, the better.
 - CLAP: cosine similarity of audio waveform and text caption embeddings
 - CCS: fraction of musical concepts present in a generated music sample from the text prompt, determined by an Audio-Language Model (ALM)
- Subjective Evaluation: listeners evaluate a generated music sample on a scale of 1-5 based on its musicality, sound quality, and adherence to the prompt

II. Participation Details and Results

- Efficiency track (<500M trainable parameters in core generative model)

FAD (lower is better)	CLAP (higher is better)	CCS (higher is better)
0.577	0.338	0.863

- **1st** in CLAP score, **2nd** in CCS score during objective evaluation of all submissions
- 2nd in efficiency track after both objective and subjective evaluation

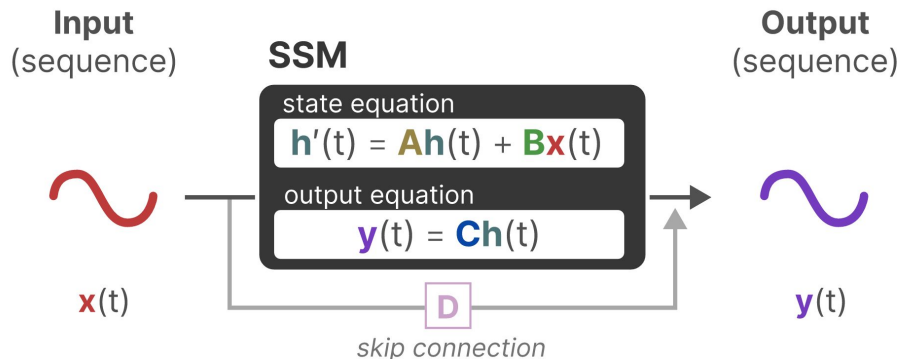
III. Architecture

Core Model

- Transformers have quadratic computational complexity wrt sequence length due to attention mechanism
 - Also tend to have large parameter counts, expensive to train
 - Alternative architectures: evaluated FNet, Perceiver, and Mamba SSM
- Mamba SSM (state-space model) appeared to be the most promising

Core Model

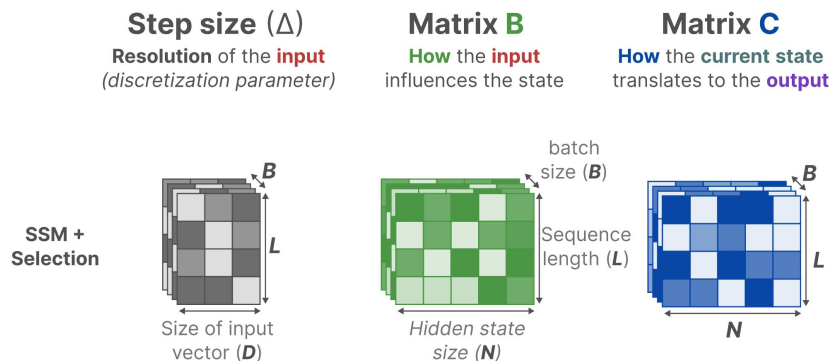
- State-Space Model (SSM): predicts the next state of a model based on the current state via a hidden state
 - Very similar to recurrent neural networks



- Setup gives advantages of RNN (linear complexity) but still suffers from memory “loss”, also not parallelizable in training

Core Model

- Solution (Gu and Dao, 2024): make the matrices A, B, C dependent on input (a “selective” SSM)!



- Using some sophisticated CUDA/GPU programming, Gu and Dao created a “hardware-aware” algorithm to speed up computation

Generative Algorithm

- Flow matching with classifier-free guidance
 - We use standard linear interpolation between the clean data and random noise at time t and regress against the straight line difference:

Suppose $v_\theta(x_t, t, c)$ is the output of the neural network with input data x_t , time t , and text embeddings c . Letting x be the “clean” data, $z \sim \mathcal{N}(0, 1)$, $x_t = (1 - t)z + tx$, we use standard mean-squared error for the loss

$$\mathcal{L}_{FM} = \mathbb{E} [||v_\theta(x_t, t, c) - (x - z)||_2^2]$$

for $t \in [0, 1]$

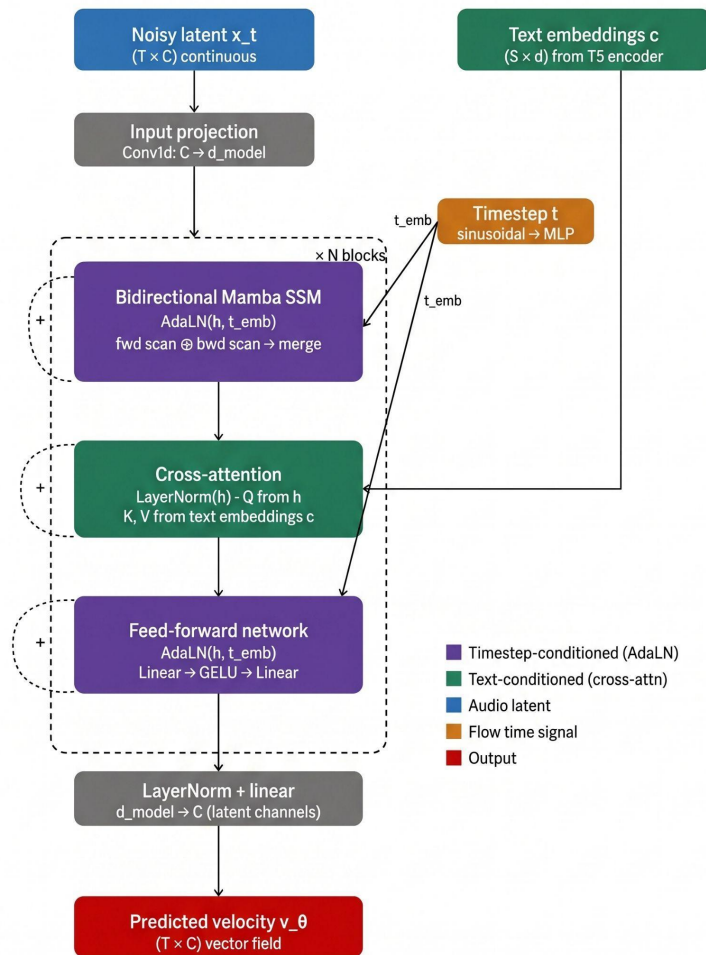
- In classifier-free guidance, drop text embeddings c randomly so model also learns the unconditional vector field for inference:

$$s_t = v_\theta(z_t, t, \emptyset) + \sigma \cdot (v_\theta(z_t, t, c) - v_\theta(z_t, t, \emptyset))$$

$$z_{t_{i+1}} = z_{t_i} + \Delta t \cdot s_{t_i}$$

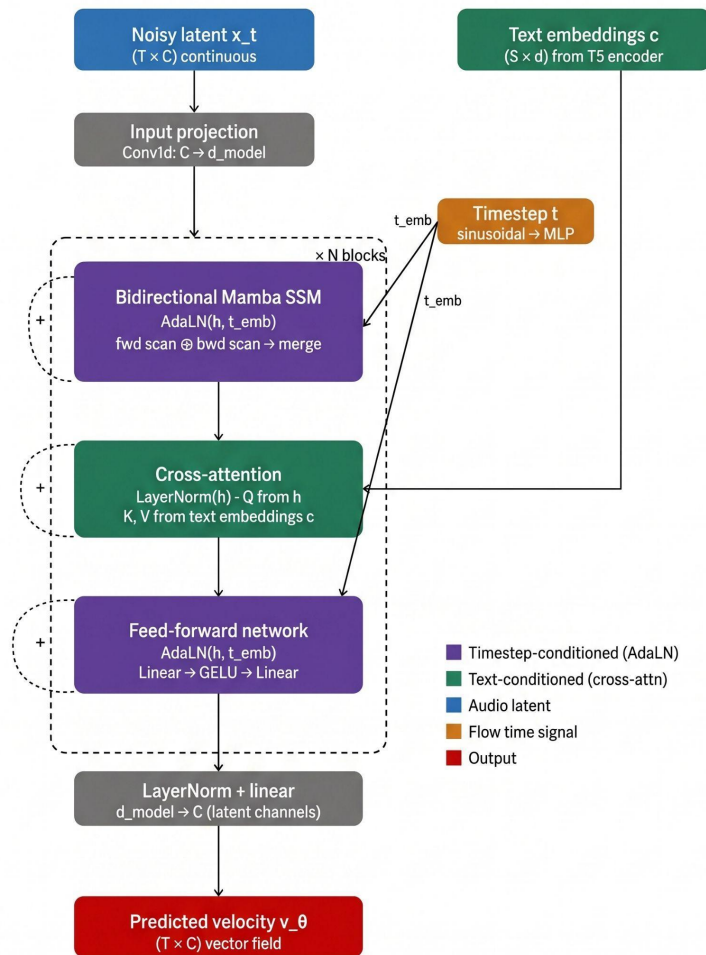
Complete Architecture

- ACE-Step-v1-3.5B Music Deep Compression Autoencoder (DCAE) for latent representation of waveform
- FLAN T5 text encoder for prompt embeddings
- $N=24$ blocks used
- $\sim 189\text{M}$ trainable parameters in total



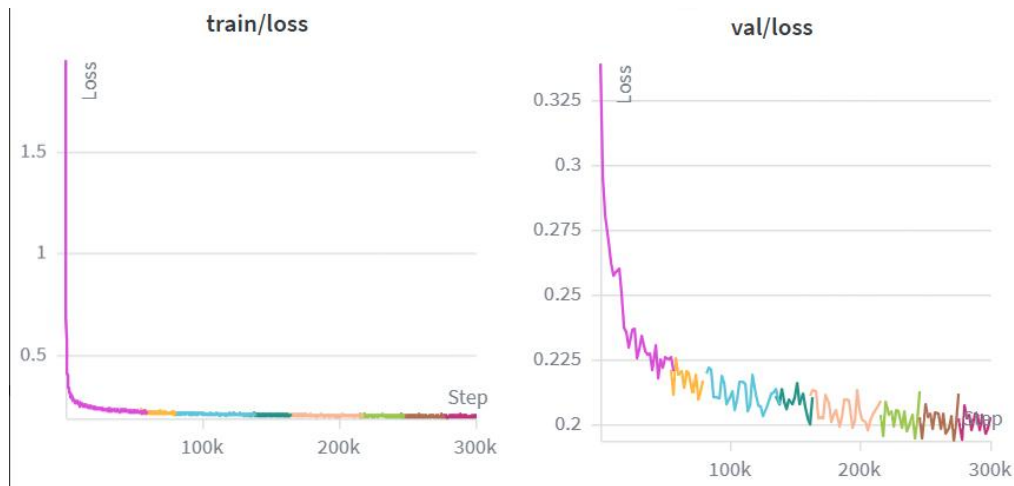
Complete Architecture

- Bidirectional Mamba: analogous to a bidirectional RNN, pass input latent in reverse to Mamba in addition to the forward direction and concatenate
 - Allows model to understand global context of the music, which is helpful for flow matching (not autoregressive)



IV. Training Details

- Hardware: 1x Nvidia RTX A5000/A30 (24 GB VRAM), AMD EPYC 7543P CPU with 8 cores, 64 GB of memory
 - Provided by UC San Diego's Data Science and Machine Learning Platform
- 300,000 training steps, ~81 hours to complete



V. Samples

Music Samples!

Prompt: “An inspiring rocknroll piece featuring a ukulele, bright and upbeat with a lively rhythm that captures the spirit of classic, joyful tunes.”



Prompt: “A mysterious club track featuring a shimmering guitar, deep sub-bass, and crisp snare rolls, with a hypnotic rhythm and subtle arpeggios that build an atmospheric tension. The guitar weaves through the mix with a dark, textured tone, enhanced by a resonant synth pad and a pulsing electronic bassline, creating an enigmatic, underground vibe perfect for late-night introspection.”

